Power Analysis for Longitudinal Clinical Trials with Run-In and Common Close Observations

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Abstract

Background. Longitudinal clinical trials often incorporate pre-randomization (run-in) and post-treatment (common close) observations to improve estimator efficiency. Closed-form power and sample-size formulas for these augmented designs under linear mixed-effects models with random slopes have remained scattered across the literature and partially specified, making applied sample-size calculation cumbersome. **Methods.** We derive closed-form power and sample-size formulas for three nested cases: (i) two run-in observations, (ii) an arbitrary number J_0 of run-in observations, and (iii) J_0 run-in plus J_2 common close observations. All derivations proceed via the Woodbury matrix identity applied to the GLS estimator under change-score parameterization. The formulas are expressed in terms of the residual variance σ^2, the random-slope variance σ_b^2, and the observation spacing t, enabling direct computation of required sample sizes. **Results.** In a numerical example using parameters from Alzheimer's disease neuroimaging trials, for which the signal-to-noise ratio $\sigma_b/\sigma \approx 1.7$ is favorable, two run-in observations reduce the required sample size by 62% (from 385 to 145 per group), and adding two common close observations raises the reduction to 82%. The magnitude of the gain is governed by σ_b/σ: across a representative range of slope variability the reduction from two run-in observations spans roughly 14% at low ratios to 76% at high ratios. The closed-form expressions allow this sensitivity analysis with respect to σ_b/σ without re-running simulation. **Conclusions.** The Woodbury-based formulas give closed-form expressions for the variance of the treatment-effect estimator under random-slopes longitudinal-trial designs with run-in and common close observations. They serve as a foundation for the AR(1) extension (Paper 2), the optimal-allocation problem (Paper 3), and the GLS-versus-ANCOVA-versus-averaged estimator comparison (Paper 4) in this compendium.

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1 Introduction

The run-in period has served multiple purposes in clinical trial design, from identifying compliant participants to washing out prior treatments[1]. A distinct and methodologically important use has emerged in trials where the primary outcome is a rate of change: estimating individual baseline trajectories that, when incorporated into the analysis model, substantially increase statistical power.

[2] established the theoretical foundation for this approach within the linear mixed effects framework. Their key insight was that when between-subject variability in rates of change is large relative to measurement error, pre-randomization observations provide valuable information about each participant's underlying trajectory, reducing the residual variance of the treatment effect estimator. Working with neuroimaging outcomes in Alzheimer's disease trials, where the ratio of slope variability to measurement error is particularly favorable, they demonstrated sample size reductions of 30–45%.

[3] extended this framework by proposing two-period linear mixed effects models that jointly model run-in and post-randomization data rather than using run-in

observations merely as covariates. Their approach yields additional power gains of up to 15% and enables unequal randomization ratios without power loss, since run-in observations effectively contribute placebo-equivalent information for all participants.

A parallel literature has developed closed-form variance and sample size formulae for the random coefficient regression model without a run-in phase.[4] derived the variance of the constrained longitudinal data analysis estimator and its relation to analysis of covariance, and[5] established efficiency conditions under which the two approaches coincide.[6] generalized these results to random coefficient regression models with monotone missing data and variable follow-up, providing variance, power, and sample size expressions illustrated with data from the Alzheimer's Disease Neuroimaging Initiative. These formulae apply to designs with $J_0 = 0$ run-in observations and an arbitrary post-randomization visit schedule, and serve as the baseline case that we extend here.

The CRAN reference implementation of the $J_0 = 0$ formulae is the `longpower` package[7], which provides the routines `edland.linear.power()`, `diggle.linear.power()`, and `liu.liang.linear.power()` for slope-comparison trials with random intercepts and slopes[8–10]. A complementary implementation is the Stata package `slopepower`[11], which obtains power for slope-comparison trials by simulating from a fitted mixed model rather than by evaluating a closed form; it addresses the same design question but does not cover run-in or common close phases, and its simulation route does not expose the variance as an explicit function of the design counts. We show in Section 3.3 that the closed-form result derived below reduces to the Ard-Edland slope-variance formula implemented in `edland.linear.power()` at the boundary $J_0 = J_2 = 0$, and to the Diggle compound-symmetry formula when additionally $\sigma_b^2 = 0$. The generalization developed here therefore embeds the existing `longpower` formulae as the baseline case against which the efficiency gains from run-in and common close observations are measured.

Despite these advances, no general closed-form power formula has been derived that simultaneously accounts for an arbitrary number of run-in observations and a separate set of common close observations collected after treatment cessation. The common close phase, in which all participants are observed off-treatment following the randomized period, provides additional information about individual trajectories at no cost to the treatment comparison. Post-treatment observation periods appear in delayed-start and randomized withdrawal designs aimed at distinguishing symptomatic from disease-modifying effects[12], but the contribution of the common close to statistical efficiency within the GLS framework has not been formally characterized. The delayed-start design[13] is the closest relative: it also observes participants after the initial randomized period, but it does so in order to test whether an early-treatment group retains an advantage over a late-start group, so the post-randomization phase carries its own treatment contrast and its own estimand. The common close considered here serves the different purpose of sharpening the estimate of each participant's underlying slope, with all participants off treatment and no contrast attached to the phase itself. A formal comparison of the two estimands, and of the assumptions each requires about post-cessation trajectories, is beyond the scope of this paper.

We present three results. First, we derive a simple closed-form variance formula for the treatment effect estimator when exactly two run-in observations precede randomization (Section 3.2). Second, we generalize this formula to an arbitrary number J_0 of run-in observations (Section 3.3). Third, we further extend to include J_2 common close observations collected after treatment cessation (Section 3.4). All derivations proceed from the generalized least squares estimator under the Woodbury matrix identity, yielding expressions in terms of the residual variance σ^2, the random slope variance σ_b^2, and the observation spacing t. Section 5 provides numerical examples using parameters estimated from Alzheimer's disease neuroimaging studies.

2 Model

2.1 General formulation

Consider a two-arm longitudinal clinical trial in which N participants are observed at equally spaced time points across three phases: a run-in period (pre-randomization), a treatment period (post-randomization), and an optional common close period (post-treatment). Let Y_{ij} denote the outcome for participant i at time point j. The observation schedule is:

- **Run-in:** $j = -J_0, -J_0 + 1, \ldots, -1$ (all participants untreated)
- **Randomization visit:** $j = 0$ (baseline)
- **Treatment period:** $j = 1, 2, \ldots, J_1$ (participants randomized to treatment or placebo)
- **Common close:** $j = J_1 + 1, J_1 + 2, \ldots, J_1 + J_2$ (all participants off-treatment)

Throughout, the phase counts (J_0, J_1, J_2) correspond one-to-one to the arguments `r` (run-in), `k` (post-randomization), and `f` (common close) of the accompanying `runinpower` package functions used in the code chunks below.

We model the outcome as

$$Y_{ij} = \alpha + a_i + (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i)t_j + u_{ij} + \varepsilon_{ij} \tag{1}$$

where α is the overall intercept, $a_i \sim N(0, \sigma_a^2)$ is a random intercept, $b_i \sim N(0, \sigma_b^2)$ is a random slope, $u_{ij} \sim N(0, \sigma_u^2)$ are within-subject between-visit random effects independent across visits, and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ are independent between-scan measurement errors. This three-component within-subject variance structure follows[2] (their Eq. 17). Because u_{ij} and ε_{ij} are both independent across visits, under a single scan per visit they enter every formula below only through their sum. We therefore write $w_{ij} = u_{ij} + \varepsilon_{ij}$ for the total within-visit residual, with

$$\sigma^2 \;\equiv\; \mathrm{Var}(w_{ij}) \;=\; \sigma_u^2 + \sigma_\varepsilon^2. \tag{2}$$

The two within-visit components are separately identifiable only when a visit carries replicate scans; for the single-scan designs treated here only their sum σ^2 enters the variance of the treatment-effect estimator. The phase indicator is

$$h_j = \begin{cases} 0 & \text{if } j \le 0 \quad \text{(run-in)} \\ 1 & \text{if } j > 0 \quad \text{(post-randomization)} \end{cases}$$

and the treatment group indicator is

$$g_i = \begin{cases} 1 & i = 1, \dots, m \quad \text{(treatment)} \\ 0 & i = m+1, \dots, N \quad \text{(placebo)} \end{cases}$$

The fixed effects are: δ, the rate of change during the run-in period (common to all participants); β, the rate of change during the post-randomization period for the placebo group; and γ, the treatment effect on the rate of change. We assume equally spaced observations with interval t, so $t_j = jt$.

For the common close phase ($j > J_1$), both treatment and placebo groups revert to a common off-treatment slope. We model this by setting $g_i = 0$ for all participants during the common close, so that the slope for both groups is $\beta + b_i$ in this phase. This is equivalent to replacing the treatment indicator with $g_i \cdot I(1 \le j \le J_1)$ in model (1).

2.2 Change-score parameterization

Following[2], we work with change scores relative to the randomization visit:

$$c_{ij} = Y_{ij} - Y_{i0} \tag{3}$$

This eliminates the intercept α and the random intercept a_i. We note that the change-score approach is algebraically equivalent to constrained longitudinal data analysis (cLDA) under compound symmetry, where the baseline is included in the response vector with a common-mean constraint across treatment groups[14,15]. The variance formulas derived below therefore apply to both change-score and cLDA analyses of run-in designs. Because the change-score eliminates a_i, the resulting formulas depend on only two variance components (σ^2, σ_b^2) rather than the three $(\sigma^2, \sigma_a^2, \sigma_b^2)$ required by the raw-outcome model.[16] show that ignoring the intercept-slope correlation in the three-component model can produce anticonservative sample sizes; the change-score approach avoids this issue by construction, at the cost of discarding information about σ_a^2.

The change-score formulation yields

$$c_{ij} = (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i)t_j + (w_{ij} - w_{i0}) \tag{4}$$

for $j \neq 0$. The change-score vector for participant i has dimension $J = J_0 + J_1 + J_2$, encompassing all three phases.

2.3 Covariance structure

The marginal covariance of the change-score vector is

$$\Sigma = R + ZGZ' \tag{5}$$

where $G = \sigma_b^2$ (scalar, since we have a single random slope), $Z = t \cdot \mathbf{1}_J$ is the $J \times 1$ random effects design vector, and R is the $J \times J$ residual covariance of the differenced within-visit residuals $(w_{ij} - w_{i0})$.

The residual covariance has the structure

$$R_{jl} = \mathrm{Cov}(w_{ij} - w_{i0},\ w_{il} - w_{i0}) = \begin{cases} 2\sigma^2 & \text{if } j = l \\ \sigma^2 & \text{if } j \neq l \end{cases} \tag{6}$$

since $w_{ij} = u_{ij} + \varepsilon_{ij}$ are independent across visits with common variance $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2$, and the shared baseline residual w_{i0} induces positive correlation among all change scores. Thus $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}'_J)$. Both within-visit components enter R identically because u_{i0} and ε_{i0} are differenced out of every change score in the same way; this is why only their sum σ^2 is identified from single-scan designs.

3 Variance of the Treatment Effect Estimator

3.1 GLS estimation and the Woodbury identity

The generalized least squares estimator of the fixed effects is

$$\hat{\beta} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c} \tag{7}$$

with $\mathrm{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$, where X is the fixed effects design matrix and $\mathbf{c}$ is the stacked change-score vector. Our target is $\mathrm{Var}(\hat{\gamma})$, the variance of the treatment effect estimator, which is the diagonal element of $(X'\Sigma^{-1}X)^{-1}$ corresponding to γ.

The key computational tool is the Woodbury matrix identity. Since $\Sigma = R + ZGZ'$ with scalar $G = \sigma_b^2$:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z\left(G^{-1} + Z'R^{-1}Z\right)^{-1}Z'R^{-1} \tag{8}$$

This avoids direct inversion of the $J \times J$ matrix Σ and enables closed-form results because R^{-1} has a known simple structure.

From $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$, the Sherman-Morrison formula gives

$$R^{-1} = \frac{1}{\sigma^2}\left(\mathbf{I}_J - \frac{1}{J+1}\mathbf{1}_J\mathbf{1}_J'\right) \tag{9}$$

With $Z = t \cdot \mathbf{1}_J$, we obtain

$$Z'R^{-1}Z = \frac{t^2}{\sigma^2}\left(J - \frac{J^2}{J+1}\right) = \frac{J\,t^2}{\sigma^2(J+1)} \tag{10}$$

and

$$\left(G^{-1} + Z'R^{-1}Z\right)^{-1} = \frac{\sigma^2\sigma_b^2(J+1)}{\sigma^2(J+1) + J\,t^2\sigma_b^2} \tag{11}$$

Let $a = \sigma^2$ and $b = \sigma_b^2$ for conciseness. Define

$$\eta_J = \frac{a(J+1) + J\,t^2 b}{a(J+1)} = 1 + \frac{J\,t^2 b}{a(J+1)} \tag{12}$$

so that $(G^{-1} + Z'R^{-1}Z)^{-1} = b/\eta_J$.

3.1.1 Per-participant information, compact form

Let X_i, Z_i, and R_i denote the per-participant fixed effects design, random effects design, and residual covariance, and define the scalar

$$\mathcal{J}_i = Z_i'R_i^{-1}Z_i + G^{-1} \tag{13}$$

By (8), the per-participant contribution to the GLS information matrix is

$$X_i'\Sigma_i^{-1}X_i = X_i'R_i^{-1}X_i - \frac{(X_i'R_i^{-1}Z_i)(Z_i'R_i^{-1}X_i)}{\mathcal{J}_i} \tag{14}$$

and the total information is $X'\Sigma^{-1}X = \sum_i X_i'\Sigma_i^{-1}X_i$, with $\mathrm{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$. Two features of this form are used later: (i) under a single random slope, $\mathcal{J}_i$ is a scalar so (14) is a rank-one correction of the OLS-style information $X_i'R_i^{-1}X_i$; (ii) because per-participant contributions add, heterogeneous designs (varying J_0, J_2, or dropout patterns) are accommodated by summing (14) over the relevant strata.

3.2 Case 1: Standard design, no run-in ($J_0 = 0,\ J_1 = 2,\ J_2 = 0$)

This case reproduces the result in[2], Section 3.1.1, and is the $J_0 = 0$ baseline of the random coefficient regression power formulae of[4] and[6]. It serves as verification for our framework.

With $J_0 = 0$ and $J_1 = 2$, there are $J = 2$ change scores $(c_{i,1}, c_{i,2})$ at post-randomization times $(t, 2t)$. Since there is no run-in, the model has two fixed effects: the placebo slope β and the treatment effect γ.

The design matrices are:

$$X_i^{(\mathrm{trt})} = t\begin{pmatrix}1 & 1\\ 2 & 2\end{pmatrix}, \qquad X_i^{(\mathrm{plc})} = t\begin{pmatrix}1 & 0\\ 2 & 0\end{pmatrix}$$

The residual covariance is $R = a\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}$ with $R^{-1} = \frac{1}{3a}\begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}$, and the random effects design is $Z = t(1,\, 2)'$.

Applying the Woodbury identity with $J = 2$:

$$Z'R^{-1}Z = \frac{t^2}{3a}(1,\,2)\begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}\begin{pmatrix}1\\ 2\end{pmatrix} = \frac{t^2}{3a}(1,\,2)(0,\,3)' = \frac{6t^2}{3a} = \frac{2t^2}{a} \tag{15}$$

$$(G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a + 2bt^2} \tag{16}$$

After summing over one treatment and one placebo participant, the information matrix is

$$X'\Sigma^{-1}X = \frac{2t^2}{a + 2bt^2}\begin{pmatrix}2 & 1\\ 1 & 1\end{pmatrix} \tag{17}$$

and its inverse is

$$\mathrm{Var}(\hat{\beta}) = \frac{a + 2bt^2}{2t^2}\begin{pmatrix}1 & -1\\ -1 & 2\end{pmatrix} \tag{18}$$

The variance of the treatment effect estimator (per participant pair) is the $(2, 2)$ element:

$$\boxed{\mathrm{Var}(\hat{\gamma})\big|_{J_0=0, J_1=2} = \frac{a + 2bt^2}{t^2} = \frac{\sigma^2 + 2\sigma_b^2 t^2}{t^2}} \tag{19}$$

This agrees with Frost et al. (2008), Section 3.1.1. For m participants per group: $\mathrm{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/(mt^2)$. This serves as the baseline against which the run-in designs are compared.

3.3 Case 2: Two run-in observations ($J_0 = 2,\ J_1 = 1,\ J_2 = 0$)

We now have $J = 3$ change scores $(c_{i,-2},\, c_{i,-1},\, c_{i,1})$ at times $(-2t,\, -t,\, t)$, the baseline visit at time 0 having been differenced away. The fixed effects design matrix separates the run-in slope δ from the post-randomization slopes:

$$X_i^{(\mathrm{trt})} = \begin{pmatrix}-2t & 0 & 0\\ -t & 0 & 0\\ 0 & t & t\end{pmatrix}, \qquad X_i^{(\mathrm{plc})} = \begin{pmatrix}-2t & 0 & 0\\ -t & 0 & 0\\ 0 & t & 0\end{pmatrix}$$

Since the random slope applies uniformly to all time points, $Z = t(-2,\, -1,\, 1)'$ for the random effect b_i. We note that the change scores at times $-2t$ and $-t$ relative to baseline at 0 yield $c_{i,-2} = Y_{i,-2} - Y_{i,0}$ and $c_{i,-1} = Y_{i,-1} - Y_{i,0}$.

More carefully, under model (1):

$$\begin{aligned} c_{i,-2} &= -2(\delta + b_i)\,t + (w_{i,-2} - w_{i,0}) \\ c_{i,-1} &= -(\delta + b_i)\,t + (w_{i,-1} - w_{i,0}) \\ c_{i,1} &= (\beta + \gamma g_i + b_i)\,t + (w_{i,1} - w_{i,0}) \end{aligned} \tag{20}$$

Thus the design matrices for the fixed effects $(\delta,\, \beta,\, \gamma)'$ are:

$$X_i^{(\mathrm{trt})} = t\begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad X_i^{(\mathrm{plc})} = t\begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

and the random effects design is $Z_i = t(-2,\, -1,\, 1)'$.

The residual covariance is

$$R = a\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = a(\mathbf{I}_3 + \mathbf{1}_3\mathbf{1}_3') \tag{21}$$

with inverse

$$R^{-1} = \frac{1}{a}\left(\mathbf{I}_3 - \frac{1}{4}\mathbf{1}_3\mathbf{1}_3'\right) = \frac{1}{4a}\begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \tag{22}$$

Now we compute the Woodbury components. With $Z = t(-2,\, -1,\, 1)'$:

$$Z'R^{-1}Z = \frac{t^2}{4a}(3\cdot 4 + 3\cdot 1 + 3\cdot 1 - 2(-1)(-2)(-1) - 2(-1)(-2)(1) - 2(-1)(-1)(1)) \tag{23}$$

Computing directly: $Z'R^{-1}Z = \frac{t^2}{4a}(-2,\, -1,\, 1)\begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}(-2,\, -1,\, 1)'$

$$R^{-1}Z = \frac{t}{4a}\begin{pmatrix} 3(-2) + (-1)(-1) + (-1)(1) \\ (-1)(-2) + 3(-1) + (-1)(1) \\ (-1)(-2) + (-1)(-1) + 3(1) \end{pmatrix} = \frac{t}{4a}\begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix}$$

so $Z'R^{-1}Z = \frac{t^2}{4a}[(-2)(-6) + (-1)(-2) + (1)(6)] = \frac{t^2}{4a}(12 + 2 + 6) = \frac{20t^2}{4a} = \frac{5t^2}{a}$.

Alternatively, we note $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-2, -1, 1)'$, so $Z'R^{-1}Z = \frac{t^2}{a}(\mathbf{d}'\mathbf{d} - \frac{1}{4}(\mathbf{1}'\mathbf{d})^2) = \frac{t^2}{a}(6 - \frac{4}{4}) = \frac{5t^2}{a}$.

Thus

$$H \equiv (G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a + 5bt^2} \tag{24}$$

For the information matrix, we compute $X'R^{-1}X$ summed over treatment and placebo groups. For a single treatment participant:

$$X_i^{(\mathrm{trt})\prime} R^{-1} X_i^{(\mathrm{trt})} = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

Computing the product:

$$X_i^{(\mathrm{trt})\prime} R^{-1} X_i^{(\mathrm{trt})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix} \tag{25}$$

For placebo (whose γ column is zero):

$$X_i^{(\mathrm{plc})\prime} R^{-1} X_i^{(\mathrm{plc})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{26}$$

Summing (one participant per group for the derivation, then multiplying by m):

$$X'R^{-1}X = \frac{t^2}{4a} \begin{pmatrix} 22 & 6 & 3 \\ 6 & 6 & 3 \\ 3 & 3 & 3 \end{pmatrix} \tag{27}$$

For the Woodbury correction, we need $X'R^{-1}Z$. For treatment:

$$X_i^{(\mathrm{trt})\prime} R^{-1} Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix}$$

For placebo:

$$X_i^{(\mathrm{plc})\prime} R^{-1} Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix}$$

The Woodbury correction requires the per-subject outer products (not the outer product of the sum):

$$W = \sum_g X_i^{(g)\prime} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)}$$

$$= \frac{bt^4}{16a(a+5bt^2)} \left(\begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix} (14,6,6) + \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix} (14,6,0) \right)$$

$$= \frac{bt^4}{16a(a+5bt^2)} \begin{pmatrix} 392 & 168 & 84 \\ 168 & 72 & 36 \\ 84 & 36 & 36 \end{pmatrix} \tag{28}$$

Therefore

$$X'\Sigma^{-1}X = X'R^{-1}X - W \tag{29}$$

To find $\mathrm{Var}(\hat{\gamma})$, we need the $(3,3)$ element of $(X'\Sigma^{-1}X)^{-1}$. We compute this numerically in the R code below and verify against the closed-form result.

Carrying through the algebra, define $D = a + 5bt^2 = \sigma^2 + 5\sigma_b^2 t^2$ and write $X'\Sigma^{-1}X = \frac{t^2}{16aD} A$. The matrix A has $(3,3)$ minor $1536\,aD$ and determinant $9216\,aD\,(a+2bt^2)$. Hence the $(3,3)$ cofactor of $X'\Sigma^{-1}X$ is $1536aD \cdot t^4/(16aD)^2 = 6t^4/(aD)$, and its determinant is $9216\,aD\,(a+2bt^2) \cdot t^6/(16aD)^3$. The ratio gives:

$$\boxed{\mathrm{Var}(\hat{\gamma})\big|_{J_0=2,J_1=1} = \frac{8\sigma^2(\sigma^2+5\sigma_b^2t^2)}{3t^2(\sigma^2+2\sigma_b^2t^2)}} \tag{30}$$

Unlike the Frost formula (19), the run-in variance depends on σ^2 both in the numerator and denominator, reflecting the additional information extracted from pre-randomization observations.

3.4 General formula: J_0 run-in observations ($J_2 = 0$)

We now derive the variance for arbitrary J_0 run-in observations with J_1 post-randomization observations and no common close ($J_2 = 0$). The total number of change scores is $J = J_0 + J_1$.

The change scores are

$$\begin{aligned} c_{i,-s} &= -s(\delta + b_i)\,t + (w_{i,-s} - w_{i0}) && s = 1, \dots, J_0 \\ c_{i,l} &= l(\beta + \gamma g_i + b_i)\,t + (w_{i,l} - w_{i0}) && l = 1, \dots, J_1 \end{aligned} \tag{31}$$

The fixed effects design matrix has columns for (δ, β, γ). Let $\mathbf{d}_{J_0} = (-J_0, -J_0+1, \dots, -1)' \cdot t$ be the run-in time vector and $\mathbf{d}_{J_1} = (1, 2, \dots, J_1)' \cdot t$ be the post-randomization time vector. Then:

$$X_i^{(\text{trt})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{d}_{J_1} \end{pmatrix}, \qquad X_i^{(\text{plc})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{0} \end{pmatrix}$$

The random effects design vector is $Z_i = (\mathbf{d}'_{J_0}, \mathbf{d}'_{J_1})'$, i.e., the full vector of time distances from baseline.

The residual covariance remains $R = a(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}'_J)$ with inverse given by (9).

3.4.1 Computing $X'\Sigma^{-1}X$

We proceed through the Woodbury identity. Define $S_{J_0} = \sum_{s=1}^{J_0} s^2 = J_0(J_0 + 1)(2J_0 + 1)/6$ and $S_{J_1} = \sum_{l=1}^{J_1} l^2 = J_1(J_1 + 1)(2J_1 + 1)/6$.

The key intermediate quantities are:

$Z'R^{-1}Z$: With $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1)'$:

$$Z'R^{-1}Z = \frac{t^2}{a}\left[(S_{J_0} + S_{J_1}) - \frac{(\sum d_j)^2}{J+1}\right] \tag{32}$$

where $\sum d_j = -J_0(J_0 + 1)/2 + J_1(J_1 + 1)/2$.

$(G^{-1} + Z'R^{-1}Z)^{-1}$: With $Z'R^{-1}Z$ given by (32),

$$H = \left(G^{-1} + Z'R^{-1}Z\right)^{-1} = \frac{b}{1 + b\, Z'R^{-1}Z},$$

the general analogue of b/η_J in (12).

$X'R^{-1}X$: This 3×3 matrix has elements computed from the block structure of X. Letting $D_{J_0} = \sum_{s=1}^{J_0} s = J_0(J_0 + 1)/2$ and $D_{J_1} = \sum_{l=1}^{J_1} l = J_1(J_1 + 1)/2$, and using $R^{-1} = \frac{1}{a}(\mathbf{I}_J - \frac{1}{J+1}\mathbf{1}\mathbf{1}')$, the information matrix summed over one treatment and one placebo participant is:

$$X'R^{-1}X = \frac{t^2}{a}\begin{pmatrix} 2\left(S_{J_0} - \frac{D_{J_0}^2}{J+1}\right) & \frac{2D_{J_0}D_{J_1}}{J+1} & \frac{D_{J_0}D_{J_1}}{J+1} \\ \frac{2D_{J_0}D_{J_1}}{J+1} & 2\left(S_{J_1} - \frac{D_{J_1}^2}{J+1}\right) & S_{J_1} - \frac{D_{J_1}^2}{J+1} \\ \frac{D_{J_0}D_{J_1}}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} \end{pmatrix} \tag{33}$$

The factor of 2 appears on entries involving both groups, while γ-related entries lack this factor since the γ column is zero for the placebo group. The positive sign on the (δ, β) cross-term arises because the run-in and post-randomization time sums have opposite signs ($\sum d_{J_0} < 0$, $\sum d_{J_1} > 0$).

Woodbury correction: The correction is the sum of per-subject rank-one outer products:

$$W = \sum_{g \in \{\text{trt},\text{plc}\}} X_i^{(g)\prime} R^{-1}Z \cdot H \cdot Z'R^{-1}X_i^{(g)}$$

which is generally rank 2 (not rank 1), so $X'\Sigma^{-1}X = X'R^{-1}X - W$.

3.4.2 Numerical implementation

The variance $\mathrm{Var}(\hat{\gamma})$ for general J_0 and J_1 does not admit a simple closed-form expression. The formula depends on σ^2 and σ_b^2 in a ratio involving S_{J_0}, S_{J_1}, D_{J_0}, D_{J_1}, and J, and the interaction between the Woodbury correction and the information matrix produces algebraically complex expressions.

We therefore implement the computation via the matrix-based Woodbury identity in R code, which evaluates the exact variance for any configuration (J_0, J_1, J_2). The closed-form expressions for $J_0 = 0$ (19) and $J_0 = 2$ (30) serve as verification checkpoints.

3.4.3 Remark: recovery of the Edland and Diggle formulae

At $J_0 = 0$, the δ column of X is empty and the information matrix reduces to a 2×2 block on (β, γ). Carrying the Woodbury algebra through, the Sherman-Morrison cancellations collapse the Woodbury correction against the $(2,1;1,1)$ structure of $X'R^{-1}X$, and the variance of the treatment effect estimator for n participants per arm becomes

$$\mathrm{Var}(\hat{\gamma})\big|_{J_0=0,J_2=0} = \frac{2}{n}\left[\sigma_b^2 + \frac{\sigma^2}{S_{xx}}\right], \qquad S_{xx} = \sum_{j=0}^{J_1}(t_j - \bar{t})^2, \tag{34}$$

where the baseline visit at $t_0 = 0$ enters through the change-score transformation. Equation (34) is the slope-comparison formula of[8] implemented in `longpower::edland.linear.power()`. Setting additionally $\sigma_b^2 = 0$ reduces Σ to compound symmetry and recovers the formula of[9] used by `longpower::diggle.linear.power()`. The Frost formula (19) is in turn the $J_1 = 2$ specialization of (34), since $S_{xx} = 2t^2$ at that setting. The general-(J_0, J_1, J_2) result derived in the next subsection therefore embeds both `longpower` closed-form routines as strict boundary cases, with the efficiency gains quantified in Section 5 accruing entirely from the information contributed by $J_0 > 0$ or $J_2 > 0$.

3.5 General formula with common close ($J_2 \geq 0$)

We now extend to include J_2 common close observations after the treatment period. The total dimension is $J = J_0 + J_1 + J_2$.

During the common close phase (time points $J_1 + 1, \dots, J_1 + J_2$), all participants are off-treatment. The change scores are:

$$c_{i,J_1+s} = [\beta(J_1 + s) + \gamma g_i \cdot J_1 + b_i(J_1 + s)]\, t + (w_{i,J_1+s} - w_{i0}) \qquad s = 1, \dots, J_2 \tag{35}$$

The treatment effect γ enters the common close change scores only through the accumulated effect during the treatment period ($\gamma g_i \cdot J_1 \cdot t$), not through the slope during the common close. This means the γ column in the design matrix has the value $J_1 t$ for all common close observations in the treatment group.

The fixed effects design matrix becomes:

$$X_i^{(\text{trt})} = t \begin{pmatrix} -J_0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ -1 & 0 & 0 \\ 0 & 1 & 1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 & J_1 \\ 0 & J_1+1 & J_1 \\ \vdots & \vdots & \vdots \\ 0 & J_1+J_2 & J_1 \end{pmatrix}$$

The β column contains the post-randomization time for the placebo slope: $0, \dots, 0, 1, \dots, J_1, J_1+1, \dots, J_1+J_2$. The γ column is $0, \dots, 0, 1, \dots, J_1, J_1, \dots, J_1$ for treatment participants and all zeros for placebo.

The random effects design remains $Z_i = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1, J_1+1, \dots, J_1 + J_2)'$.

The residual covariance is still $R = a(\mathbf{I}_J + \mathbf{1}_J \mathbf{1}_J')$ with $J = J_0 + J_1 + J_2$.

The Woodbury computation proceeds identically, with the only changes being the dimension J, the structure of the X matrix (specifically the γ column), and the elements of Z.

We implement the general computation in R below.

4 Power and Sample Size Formulas

For a two-sided test at significance level α with power $1 - \beta_{\text{pow}}$, the required number of participants per group is

$$m = \frac{(z_{\alpha/2} + z_{1-\beta_{\text{pow}}})^2 \cdot \text{Var}_1(\hat{\gamma})}{\Delta^2} \tag{36}$$

where $\text{Var}_1(\hat{\gamma})$ denotes the per-participant-pair variance from the formulas derived above, and Δ is the minimum clinically important treatment effect on the rate of change.

Equivalently, the power for a given sample size is

$$1 - \beta_{\text{pow}} = \Phi\left(\frac{\Delta\sqrt{m}}{\sqrt{\text{Var}_1(\hat{\gamma})}} - z_{\alpha/2} \right) \tag{37}$$

The relative efficiency of a design with (J_0, J_1, J_2) observations compared to a standard design with $(0, J_1, 0)$ is

$$\mathrm{RE}(J_0, J_1, J_2) = \frac{\mathrm{Var}_1(\hat{\gamma})\big|_{0,J_1,0}}{\mathrm{Var}_1(\hat{\gamma})\big|_{J_0,J_1,J_2}} \tag{38}$$

Values greater than 1 indicate that the run-in/common close design requires fewer participants.

A practical caveat: adding a single run-in observation ($J_0 = 1$) introduces the run-in slope parameter δ into the model, increasing the fixed-effect dimension from 2 to 3. This additional parameter absorbs part of the information contributed by the run-in observation. Numerical evaluation over a wide grid of configurations ($J_1 = 2, \dots, 6$; σ_b^2/σ^2 from 10^{-4} to 10^3; t from 0.25 to 4) shows that $\mathrm{Var}(\hat{\gamma})|_{J_0=1}$ never exceeds $\mathrm{Var}(\hat{\gamma})|_{J_0=0}$, but the improvement can be negligible: at some parameter values the information from the single run-in observation is almost exactly offset by the cost of estimating δ. A single run-in observation is therefore never harmful to the precision of $\hat{\gamma}$, although the gain it provides should be quantified with the formulas above before committing to a longer trial. This is an instance of the general design principle set out by[17], that the value of an additional measurement occasion in a longitudinal trial depends jointly on the variance components and on the parameters the enlarged model must estimate, so that the number and placement of visits should be chosen by direct evaluation of the resulting precision rather than by a fixed rule.

5 Numerical Examples

5.1 Verification of closed-form results

We verify that the closed-form expressions agree with the matrix computation. The Frost et al. (2008) result corresponds to the standard design with $J_0 = 0$, $J_1 = 2$ (two post-baseline visits at times t and $2t$), which under the two-parameter model yields $\mathrm{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$. The $J_0 = 2$, $J_1 = 1$ case uses the three-parameter model.

```r
a <- 1.0
b <- 0.5
tt <- 1.0

v_frost_matrix <- var_gamma_matrix(0, 2, 0, a, b, tt)
v_frost_closed <- var_gamma_frost(a, b, tt)

v_r2_matrix <- var_gamma_matrix(2, 1, 0, a, b, tt)
v_r2_closed <- var_gamma_r2(a, b, tt)

cat('Frost (r=0, k=2) case:\n')
```

```
## Frost (r=0, k=2) case:
```

```r
cat('  Matrix:', round(v_frost_matrix, 6), '\n')
```

```
##   Matrix: 2
```

```r
cat('  Closed:', round(v_frost_closed, 6), '\n')
```

```
##   Closed: 2
```

```r
cat('  Match:',
    all.equal(v_frost_matrix, v_frost_closed), '\n\n')
```

```
##   Match: TRUE
```

```r
cat('r=2, k=1 case:\n')
```

```
## r=2, k=1 case:
```

```r
cat('  Matrix:', round(v_r2_matrix, 6), '\n')
```

```
##   Matrix: 4.666667
```

```r
cat('  Closed:', round(v_r2_closed, 6), '\n')
```

```
##   Closed: 4.666667
```

```r
cat('  Match:',
    all.equal(v_r2_matrix, v_r2_closed), '\n\n')
```

```
##   Match: TRUE
```

```r
cat('Variance comparison across designs:\n')
```

```
## Variance comparison across designs:
```

```r
for (rr in 0:4) {
  v <- var_gamma_matrix(rr, 2, 0, a, b, tt)
  cat(sprintf('  r=%d, k=2: Var = %.4f\n', rr, v))
}
```

```
##   r=0, k=2: Var = 2.0000
##   r=1, k=2: Var = 2.0000
##   r=2, k=2: Var = 1.7910
##   r=3, k=2: Var = 1.5000
##   r=4, k=2: Var = 1.2651
```

5.2 Validation of the general routine

The closed-form checkpoints above both have $J_2 = 0$, so they do not exercise the common-close path. We therefore validate the general routine in two ways. First, we compare the Woodbury computation with a direct GLS evaluation in which $\Sigma = R + \sigma_b^2 ZZ'$ is assembled and inverted without the Woodbury identity, over a grid of designs that includes configurations with $J_2 > 0$. Second, we run a seeded Monte Carlo benchmark at $(J_0, J_1, J_2) = (2, 2, 2)$: change-score vectors are simulated from the generative model, $\hat{\gamma}$ is computed by GLS in each replicate, and the empirical variance is compared with the analytic value.

```r
var_gamma_direct <- function(r, k, f, sigma2, sigma_b2,
                             t_interval) {
  p <- r + k + f
  R <- sigma2 * (diag(p) + matrix(1, p, p))
  des <- build_design(r, k, f, t_interval)
  Sigma <- R + sigma_b2 * des$Z %*% t(des$Z)
  Si <- solve(Sigma)
  info <- t(des$X_trt) %*% Si %*% des$X_trt +
    t(des$X_plc) %*% Si %*% des$X_plc
  solve(info)[des$ncol_X, des$ncol_X]
}

grid <- expand.grid(r = 0:3, k = 1:3, f = 0:3,
                    sigma_b2 = c(0.1, 1, 5),
                    t_int = c(0.5, 1, 2))
grid <- grid[grid$r + grid$k + grid$f >= 3, ]
diffs <- mapply(
  function(r, k, f, sb2, tt) {
    abs(var_gamma_matrix(r, k, f, 1.0, sb2, tt) -
          var_gamma_direct(r, k, f, 1.0, sb2, tt))
  },
  grid$r, grid$k, grid$f, grid$sigma_b2, grid$t_int
)
cat('Direct GLS vs Woodbury routine over', nrow(grid),
    'configurations (including f > 0):\n')
```

```
## Direct GLS vs Woodbury routine over 396 configurations (including f > 0):
```

```r
cat('  Maximum absolute difference:',
    format(max(diffs), digits = 3), '\n\n')
```

```
##   Maximum absolute difference: 3.38e-14
```

```r
set.seed(20260418)
r_v <- 2; k_v <- 2; f_v <- 2
s2_v <- 1; sb2_v <- 1; t_v <- 1
m_v <- 200; n_sim <- 2000
des <- build_design(r_v, k_v, f_v, t_v)
p_v <- r_v + k_v + f_v
Sigma <- s2_v * (diag(p_v) + matrix(1, p_v, p_v)) +
```

```r
  sb2_v * des$Z %*% t(des$Z)
Si <- solve(Sigma)
cR <- chol(Sigma)
info <- m_v * (t(des$X_trt) %*% Si %*% des$X_trt +
                 t(des$X_plc) %*% Si %*% des$X_plc)
theta <- c(0.2, -0.5, 0.3)
gamma_hat <- numeric(n_sim)
for (s in seq_len(n_sim)) {
  xty <- 0
  for (X in list(des$X_trt, des$X_plc)) {
    mu <- as.numeric(X %*% theta)
    Y <- matrix(rep(mu, m_v), m_v, p_v, byrow = TRUE) +
      matrix(rnorm(m_v * p_v), m_v, p_v) %*% cR
    xty <- xty + t(X) %*% Si %*% colSums(Y)
  }
  gamma_hat[s] <- solve(info, xty)[3]
}
cat('Monte Carlo at (J0, J1, J2) = (2, 2, 2), m =', m_v,
    'per group,', n_sim, 'replicates:\n')
```

```
## Monte Carlo at (J0, J1, J2) = (2, 2, 2), m = 200 per group, 2000 replicates:
```

```r
cat('  Empirical  m * Var(gamma_hat):',
    round(m_v * var(gamma_hat), 4), '\n')
```

```
##   Empirical  m * Var(gamma_hat): 1.1039
```

```r
cat('  Theoretical (Woodbury):        ',
    round(var_gamma_matrix(r_v, k_v, f_v, s2_v, sb2_v,
                           t_v), 4), '\n')
```

```
##   Theoretical (Woodbury):        1.1344
```

5.3 Sample size as a function of run-in observations

5.4 Sample size as a function of common close observations

5.5 Relative efficiency surface

5.6 Application: Alzheimer's disease imaging trial

Using variance parameters estimated from the MIRIAD dataset[2]: $\sigma_b^2 = 0.9745$ (between-subject slope variance) and a combined within-visit residual $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2 = 0.3338 + 0.0025 = 0.3363$, formed per Eq. (2) from the within-subject between-visit component ($\sigma_u^2 = 0.3338$) and the additional between-scan measurement error ($\sigma_\varepsilon^2 = 0.0025$) of Frost's Table II. We take observation interval $t = 1$ year and treatment effect $\Delta = 0.25$ (25% reduction in atrophy rate relative to

Table 1: Required sample size per group ($J_1 = 2$, $J_2 = 0$, $\Delta = 0.5$, 90% power, two-sided $\alpha = 0.05$) as a function of the number of run-in observations J_0, for three levels of between-subject slope variability.

J_0	Low	Medium	High
0	51	127	463
1	44	119	214
2	44	91	113
3	43	70	76
4	42	56	59
5	40	48	49
6	38	42	43

Table 2: Required sample size per group ($J_0 = 2$, $J_1 = 2$, $\Delta = 0.5$, 90% power) as a function of common close observations J_2.

J_2	Low	Medium	High
0	44	91	113
1	34	69	80
2	29	48	52
3	25	35	37
4	22	27	28

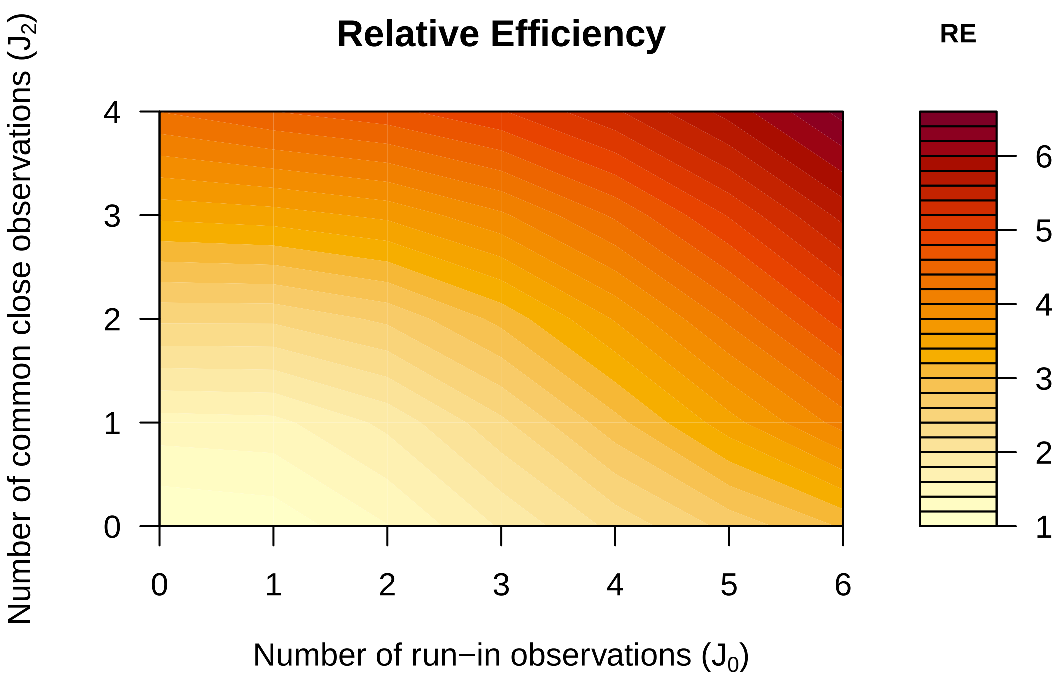

Figure 1: (#fig:fig_efficiency)Relative efficiency of run-in and common close designs compared to the standard design ($J_0 = 0$, $J_2 = 0$) with the same number of post-randomization observations ($J_1 = 2$). Parameters: $\sigma^2 = 1$, $\sigma_b^2 = 1$, $t = 1$.

Table 3: Sample size per group for AD imaging trial ($J_1 = 2$ post-randomization visits, $\Delta = 0.25$, 90% power). All $J_1 = 2$ configurations ordered by sample size; the $(0, 2, 0)$ row is the standard-design baseline.

J_0	J_1	J_2	m per group	Total visits
4	2	2	49	9
3	2	2	58	8
4	2	1	60	8
2	2	2	69	7
3	2	1	77	7
0	2	2	77	5
1	2	2	77	6
4	2	0	78	7
3	2	0	100	6
2	2	1	104	6
1	2	1	138	5
2	2	0	145	5
0	2	1	150	4
1	2	0	247	4
0	2	0	385	3

placebo rate of approximately 1.0% brain volume loss per year). Using the combined residual, the conventional design reproduces Frost's reported sample sizes (Table III); using only the 0.0025 measurement-error term, as in earlier drafts, understates the residual by two orders of magnitude and yields degenerate sample sizes.

```
## AD imaging trial sample sizes (per group):
##   Standard (r=0, k=2, f=0): 385
##   Run-in   (r=2, k=2, f=0): 145  (62% reduction)
##   Full     (r=2, k=2, f=2): 69  (82% reduction)
```

6 Discussion

The results presented here formalize and extend the framework of[2] by providing explicit variance formulas for the treatment effect estimator under designs that incorporate run-in and common close observations. Three points merit discussion.

When run-in observations are most beneficial. The efficiency gain from run-in observations depends on the ratio $\sigma_b^2 t^2/\sigma^2$. When between-subject slope variability is large relative to residual variance (as in neuroimaging outcomes), even one or two run-in observations yield substantial sample size reductions. For cognitive outcomes, where this ratio is typically less favorable, the gains are more modest and the cost of extended study duration must be weighed against the sample size reduction.

Common close as a low-cost efficiency gain. The common close phase provides information about individual trajectories without requiring participants to remain on treatment. Since treatment has ceased, both groups contribute to estimation of the underlying slope variability, effectively tightening the precision of the treatment effect estimator. The marginal benefit of each additional common close observation diminishes (as with run-in observations), but the first one or two common close visits typically provide meaningful sample size reductions, particularly when $\sigma_b^2 t^2/\sigma^2$ is large.

Relationship to the two-period model. The formulas derived here are consistent with the two-period linear mixed effects model of[3]. Their Scenario 1 (common slope across periods) corresponds to setting $\delta = \beta$ in our model, while their Scenario 2 (different slopes) corresponds to the general case. The advantage of our formulation is the explicit separation of run-in, treatment, and common close phases, enabling direct optimization of the number of observations in each phase.

Limitations. The derivations assume equally spaced observations, linear trajectories, and a random intercept plus random slope covariance structure. Extensions to unequal spacing, nonlinear progression, and more complex random effects structures (e.g., random intercept-slope correlation) are possible within the same Woodbury identity framework but yield more complex expressions that may not admit simple closed forms. The formulas also assume no dropout; in practice, the Dawson-Lagakos pattern-mixture approach[2] can be applied to adjust for anticipated attrition.

7 Conflict of interest

The author declares no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The `runinpower` R package, the four companion manuscripts of this compendium, the Mathematica symbolic-derivation scripts, and the shared bibliography are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

10 Reproducibility

This manuscript was rendered with R 4.4.x. Principal packages used are the in-package `runinpower` source (nine source files: allocation, ancova, averaged, covariance, covariance-ar1, design, power, replicated, variance), `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build. Mathematica symbolic derivations supporting the Woodbury-based variance formulas are at `analysis/scripts/`.

Random-number generator. The variance formulas are deterministic given fixed input parameters; no RNG is exercised in their evaluation. The validation chunk in the Numerical Examples section first compares the Woodbury routine with a direct GLS computation, in which Σ is assembled and inverted without the Woodbury identity, over a grid of designs that includes common-close configurations ($J_2 > 0$). It then runs a Monte Carlo benchmark at $(J_0, J_1, J_2) = (2, 2, 2)$ with 2000 replicates of $m = 200$ participants per group, setting `set.seed(20260418)` at its entry point and comparing the empirical variance of $\hat{\gamma}$ with the analytic value.

Data and code availability. The R package source is at `R/`. Mathematica scripts are at `analysis/scripts/`. The companion manuscripts are at `analysis/report/{02,03,04}-*/`. The manuscript source and shared bibliography (`../references.bib`) are at `analysis/report/01-runin-power/`.

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